CrossMEP: A Tiered Synthetic Dataset of Multi-Trade MEP Cross-Sections

A support-design problem distribution for learning-based structural support assembly synthesis

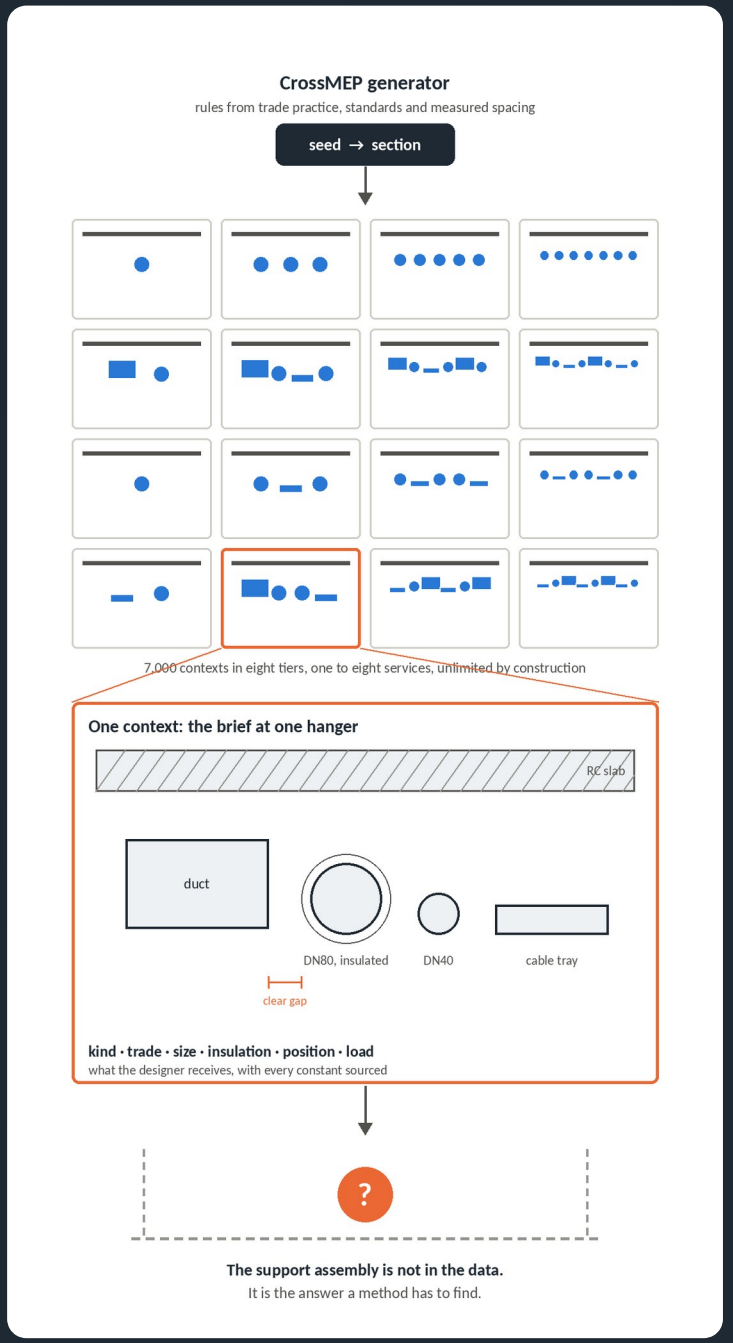
Lavinia Pedrollo [1] · **David Gvadzabia** [2] · **Torben Graeber** [3] · **Martin Fischer** [1]

[1] Stanford University, CIFE [2] Lafayette College [3] Hilti AG

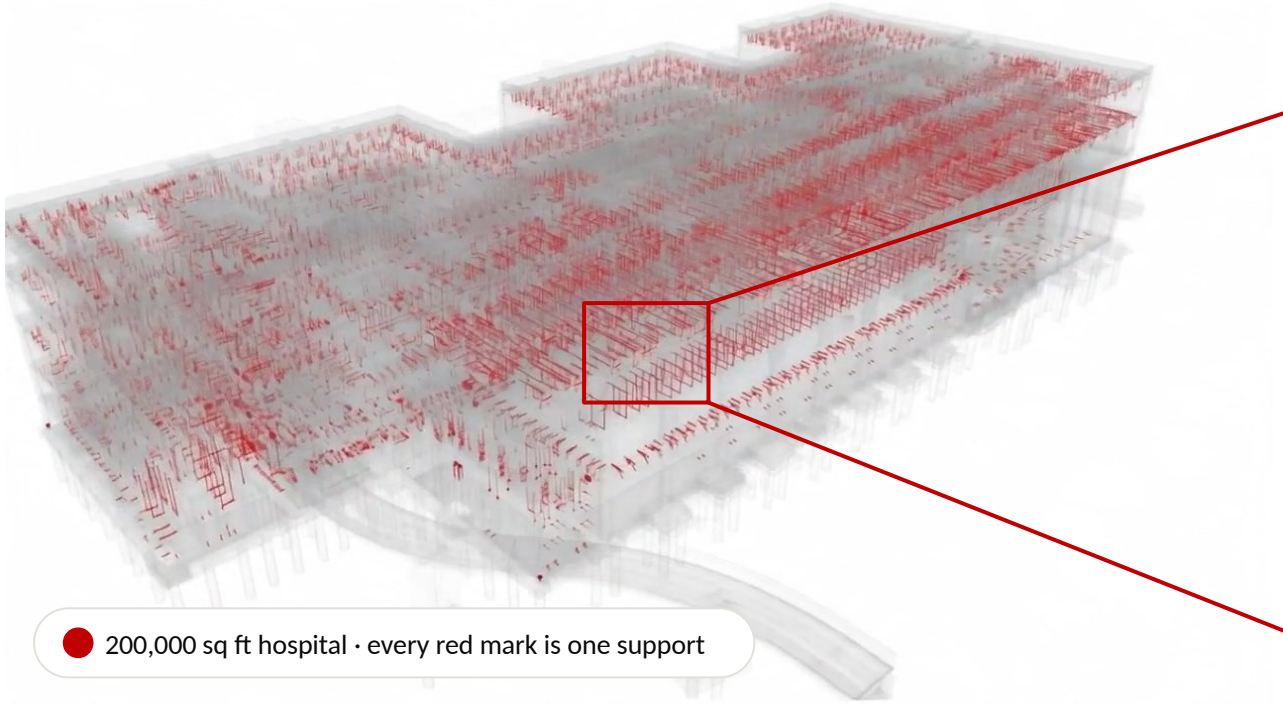
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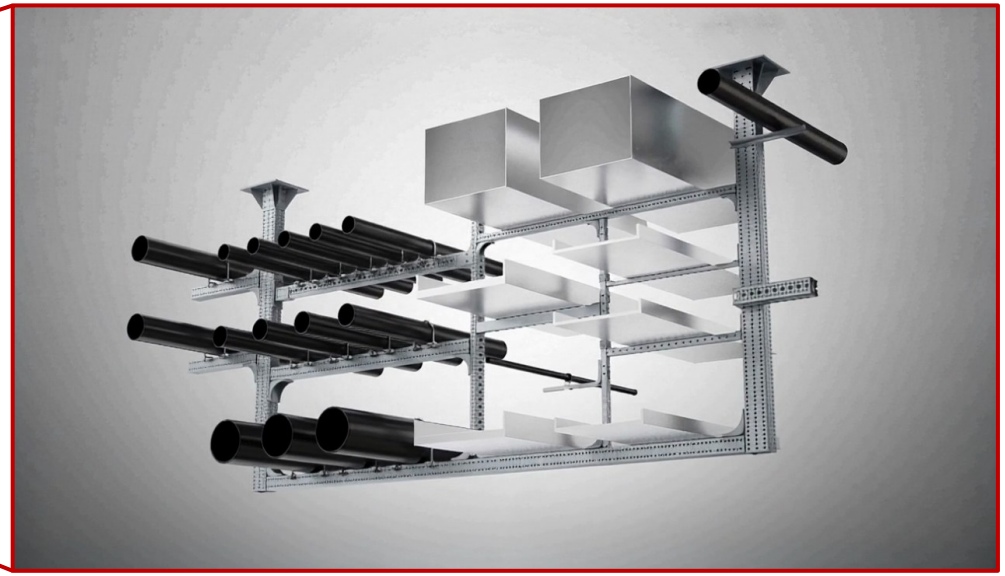
HILTI



The synthesis of structural support assemblies (SSAs) takes 25% of MEP design effort because every assembly must be designed individually



1 unit = 1 structural support assembly (SSA)



≈ 10,000
support assemblies in one hospital project, each one its own design

×

20 min – 2 h
of engineering time for each assembly, designed one by one

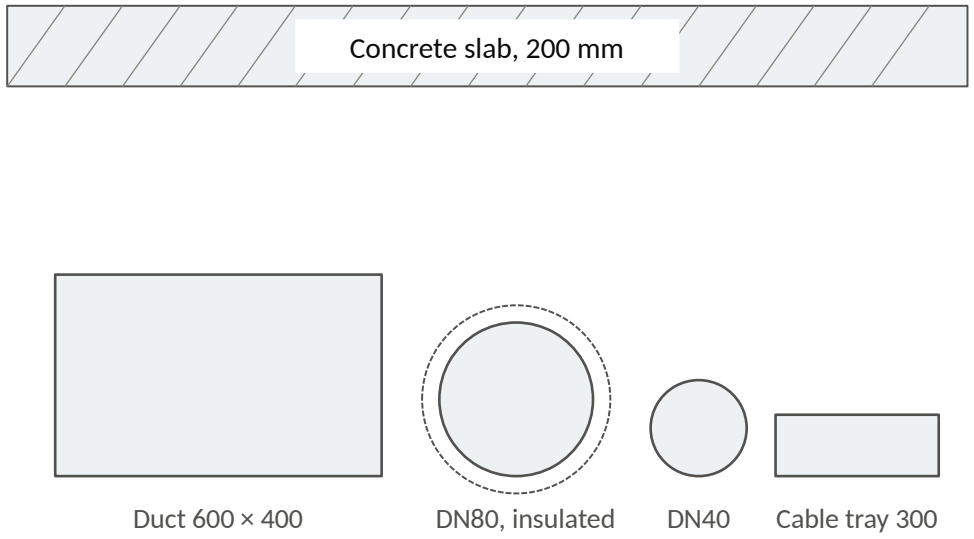
=

≈ 25 %
of MEP design effort, about $600k of engineering cost per project

Approximate practitioner estimates [1]. [1] Pedrollo, Graeber & Fischer (2026), Feasibility-aware sequential synthesis of structural support assemblies, ISARC 2026

An SSA designer turns one MEP cross-section into a feasible structural support assembly

CONTEXT · the brief

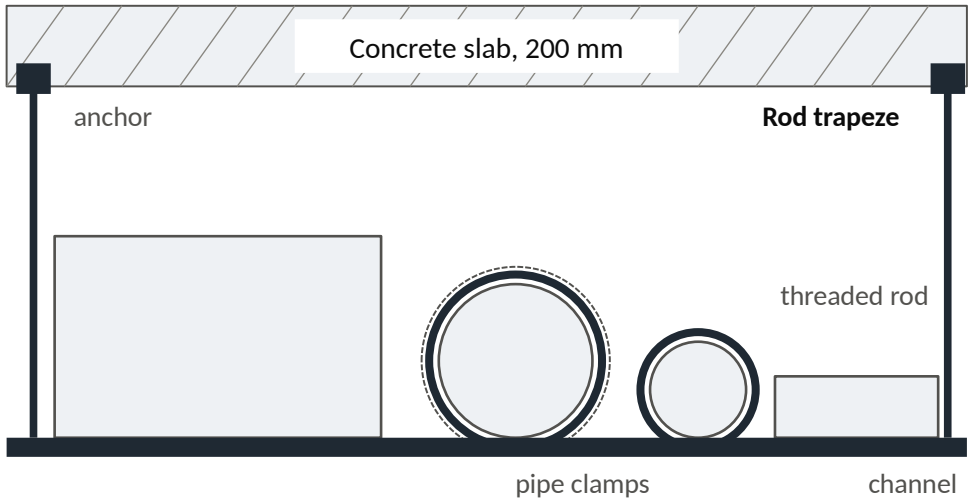


Every element with its kind, trade, size, insulation, position and load per metre. Nothing else.

SSA designer

from a catalog

FEASIBLE SSA · the answer



✓ statics ✓ anchors ✓ connectors ✓ buildable

The synthesis problem as defined at ISARC 2026 [1]. [1] Pedrollo, Graeber & Fischer (2026), Feasibility-aware sequential synthesis of structural support assemblies, ISARC 2026

Existing tools coordinate and verify modular support assemblies, but their synthesis remains manual because it relies on tacit engineering expertise and complex catalog-driven rules

Coordinate the model

tools



services routed and clash-free

Synthesize the assembly

by hand



topology and parts chosen by an engineer [1]

Verify the design

tools

✓ statics ✓ anchors ✓ connectors

statics, anchors and connectors checked

Why the middle step resists automation

Tacit judgment
the topology comes from experience

Rule explosion
constraints interact across thousands of combinations

Catalog volatility
catalogs change faster than rule systems

[1] Pedrollo, Graeber & Fischer (2026), Feasibility-aware sequential synthesis of structural support assemblies, ISARC 2026 [2] Pedrollo, Gvadzabia, Graeber & Fischer (2026), CrossMEP, CIB W78 2026 (this paper) [3] Korman, Fischer & Tatum (2003), Knowledge and reasoning for MEP coordination, J. Constr. Eng. Manage. 129(6)

A learning method needs thousands of contexts to practise on, and real projects give too few, so we generate them

How a learning method learns, round after round

pass or fail is fed back, and the method improves

1 **Context**
one section at a hanger, without the answer

2 **Learning method**
a black box that proposes a design

3 **Proposed design**
an assembly: channel, rods, clamps, anchors

4 **Check**
statics, anchors, connectors, buildability, cost

Real projects

too few, not diverse enough



A handful of sections, from one project with one trade mix, and confidential. Not enough to practise on, and a method tuned on them only fits that project.

CrossMEP generates them

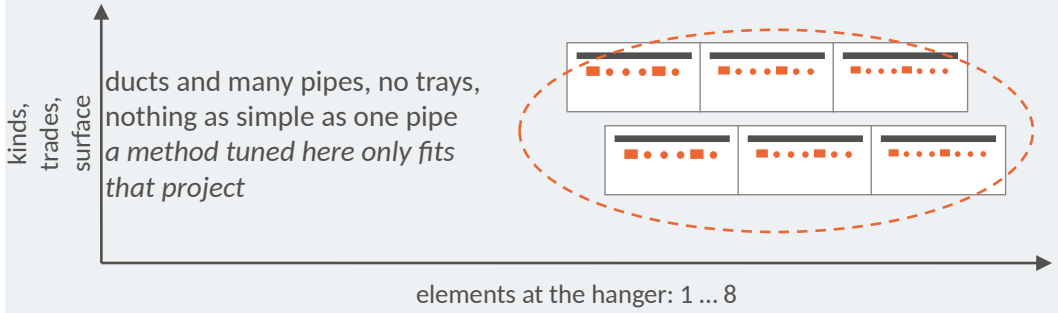
unlimited, diverse



As many contexts as a method needs: every tier from one to eight elements, kinds and surfaces varied, a new seed a new set. And one fixed benchmark, never trained on, to judge every method on.

The ninety real designs we had came from one project and filled one corner of the design space, so CrossMEP covers all of it by construction

The real designs we had ninety · one project · confidential



CrossMEP every tier · kinds and surfaces varied · unlimited



CrossMEP at a glance

7,000
contexts in four splits, on disjoint seeds

31,484
elements: pipes, trays, ducts, conduits; six trades

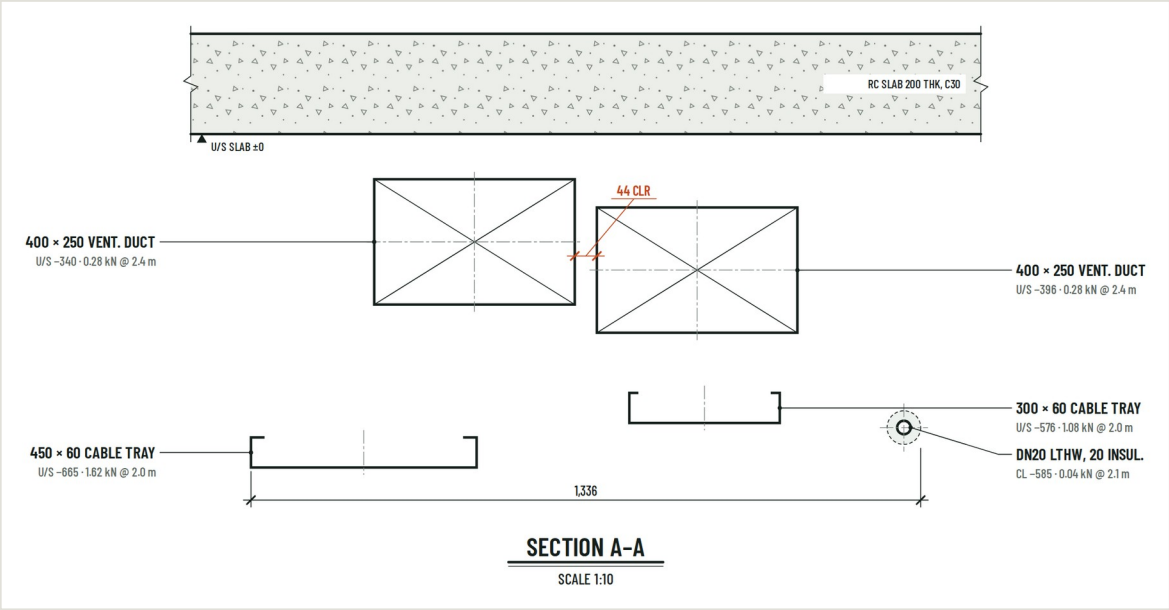
C1–C8
eight tiers by exact element count

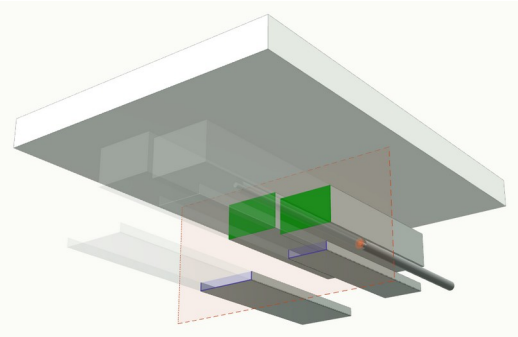
0 labels
the brief, not the answer

Traced
every constant sourced or declared

Open
data CC BY 4.0, code MIT, seeded generator

One context is exactly what a support designer receives at one hanger, without the answer





● ventilation ● electrical ● heating

What every element carries

kind | service, trade | size | insulation | position | load = kN/m × span

What the support designer is given

Element	Trade	Ins. mm	kN/m	Span	kN at support
400×250 duct	● ventilation	0	0.115	2.4 m	**0.28**
400×250 duct	● ventilation	0	0.115	2.4 m	**0.28**
450 tray	● electrical	0	0.809	2.0 m	**1.62**
300 tray	● electrical	0	0.540	2.0 m	**1.08**
DN20	● heating	20	0.019	2.1 m	**0.04**

200 mm
ceiling slab, concrete C30

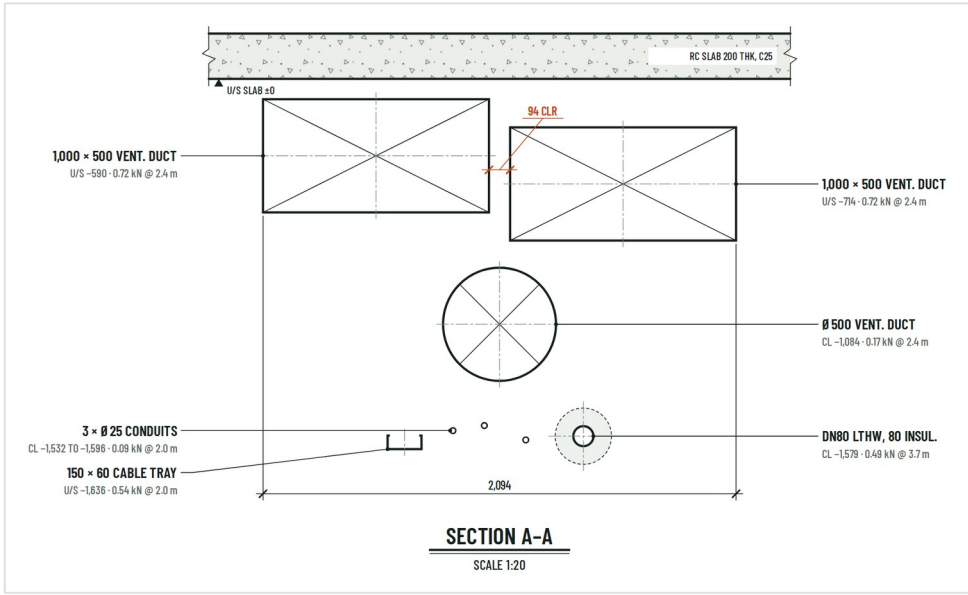
3.30 kN
total load at this support

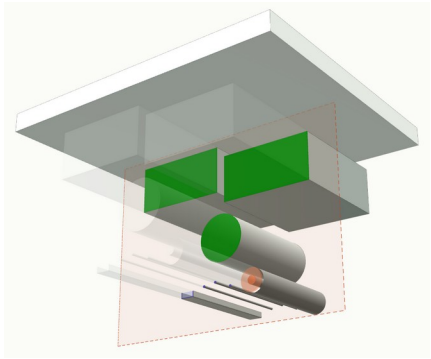
44 mm
closest clear gap, 2 rows

Not given, by design: channel, rods, clamps, anchors. No correct answer either: a feasible support depends on the catalog you build from.

Every section is built the way trades run services, with gaps measured on a real building

1. **Tier**
   Cn fixes the exact element count, n = 1 … 8
2. **Surface**
   ceiling or wall (78 / 22 %); concrete, 150–300 mm thick
3. **Services**
   grouped as trades run [3]: hot + cold, flow + return, conduit groups, 1–2 trays, ducts
4. **Rows**
   bulky first: ducts nearest the slab, then trays and conduits, then pipes
5. **Spacing**
   gaps drawn from a built project [12], at least 25 mm; electrical above wet on walls [10]
6. **Check**
   every pair 25 mm clear, rows centred, everything recorded; same seed, same file





● ventilation ● electrical ● heating

One generated context, tier C8: 8 services on 3 rows, ducts nearest the slab, then the tray and conduits, then the pipe. The gap marked on the sheet is one draw from the measured distribution; the same seed gives the same section.

[3] Korman, Fischer & Tatum (2003), Knowledge and reasoning for MEP coordination, J. Constr. Eng. Manage. 129(6) [10] IET On-Site Guide (BS 7671), conduit support spacing; BS 7671 Reg. 528.3.2 [12] buildingSMART International (2020), Duplex Apartment and Medical-Dental Clinic sample IFC files, CC BY 4.0

Every number is traced to a standard, to a measurement, or to a choice we declare as ours



Pipes DN15–150

Sizes EN 10220 / EN 10255 medium series [4]
Load steel + water × ASME B31.1 water-service span [8]



Insulation

Sizes GEG Anlage 8, heated lines [11]; 30 / 50 mm condensation control, chilled
Load adds to the size, not the load



Cable trays 150–600

Sizes IEC 61537 systems [6] in NEMA VE 1 widths [7]
Load 50 kg/m full at 300 mm × 2.0 m, below the shortest NEMA class span [7]



Ducts

Sizes EN 1505 rectangular, EN 1506 round [5]
Load manufacturer weight table with flanges × 2.4 m (SMACNA) [9]



Conduits Ø20–50

Sizes IEC 61386-1 [6], parallel groups of 2–6
Load steel tube + 40 % cable fill × 2.0 m (IET / BS 7671) [10]



Standards
what each element is and weighs: sizes, walls, spans, insulation [4]–[11]



Measured
how close neighbours sit: clear gaps and row stagger from two open buildings [12]



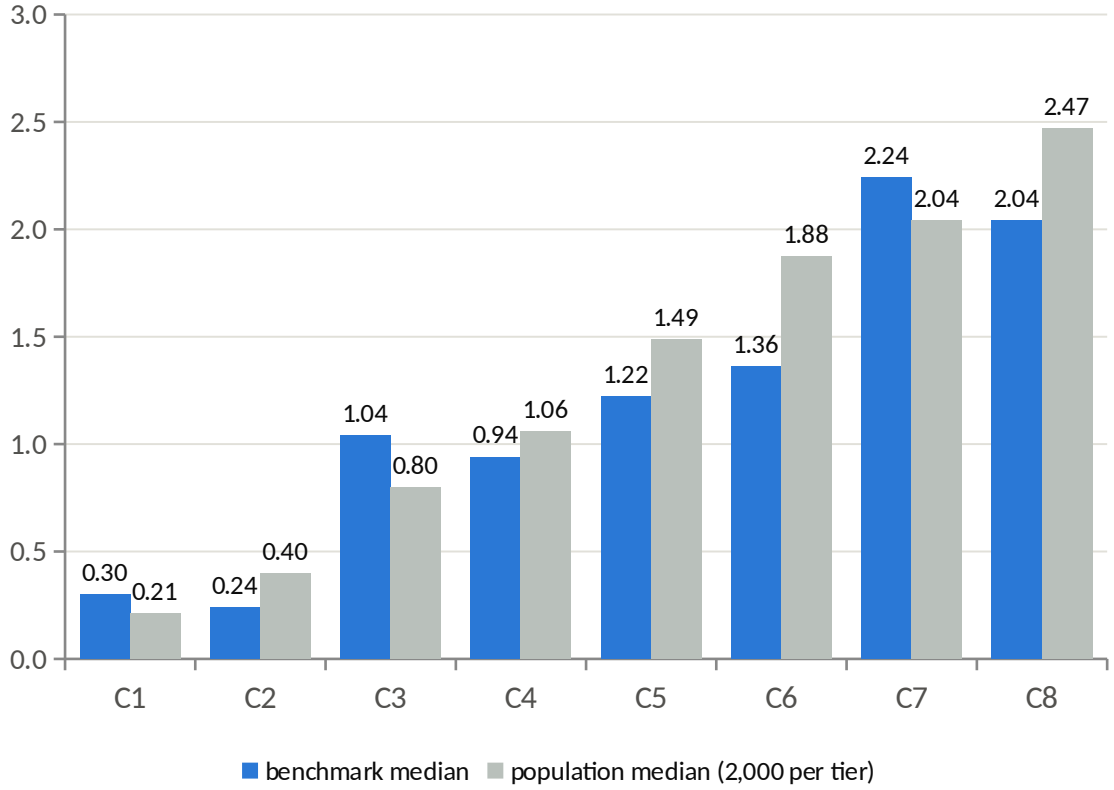
Declared
our choices, labelled as such: trade mix, ceiling / wall 78 / 22 %, number of rows

Every constant carries its source and its status, and why each standard was chosen over its alternatives is written down with it.

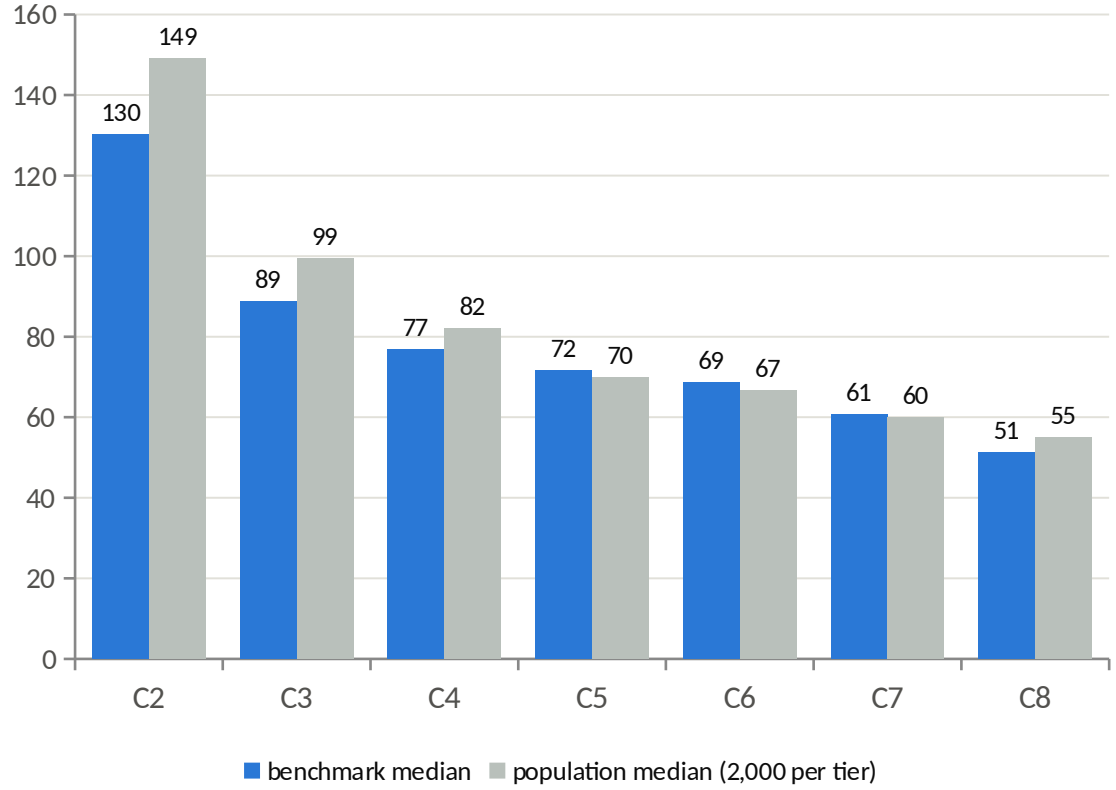
[4] EN 10220:2002, EN 10255:2004 (steel tubes: dimensions, medium series) [5] EN 1505:1997, EN 1506:2007 (sheet-metal air ducts: rectangular and circular dimensions) [6] IEC 61386-1 (conduit systems), IEC 61537 (cable tray and ladder systems) [7] NEMA VE 1 (metal cable tray systems: widths, load-class spans) [8] ASME B31.1 Table 121.5 (hanger spacing), as republished in the ASHRAE Handbook, HVAC Systems and Equipment [9] SMACNA, HVAC Duct Construction Standards, Metal and Flexible, Table 5-1 [10] IET On-Site Guide (BS 7671), conduit support spacing; BS 7671 Reg. 528.3.2 [11] GEG (Gebäudeenergiegesetz), Anlage 8 (pipe insulation thickness) [12] buildingSMART International (2020), Duplex Apartment and Medical-Dental Clinic sample IFC files, CC BY 4.0

The tier fixes the element count, and load and congestion grow with it as the rules intend

Heavier with the tier: median load at the support (kN)

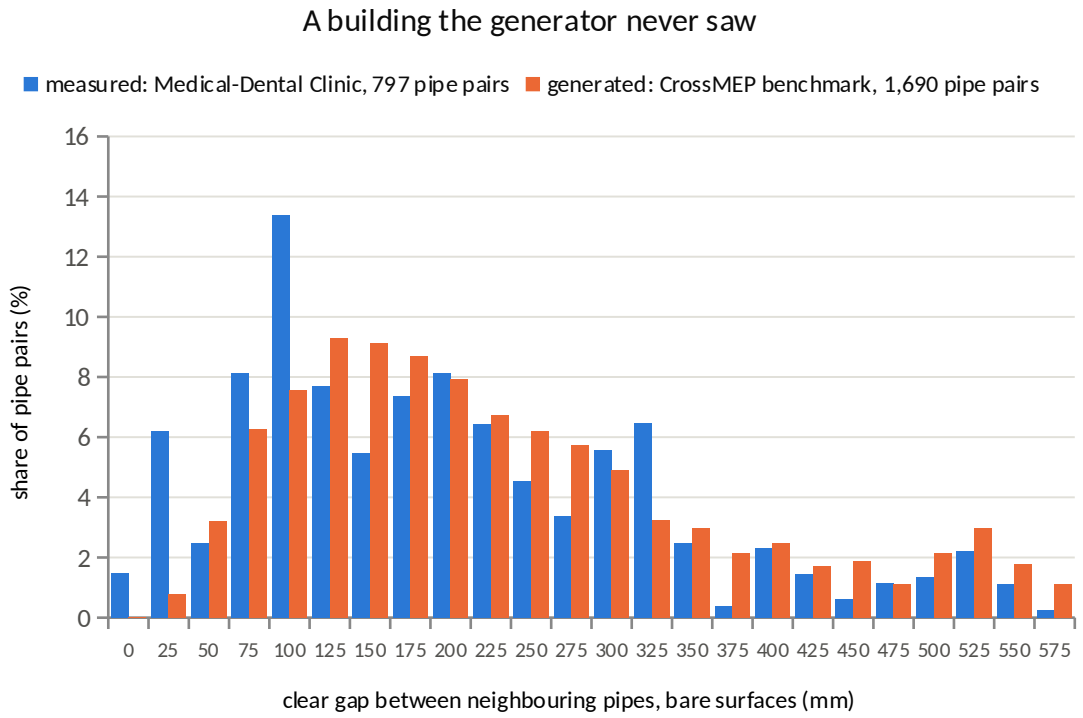


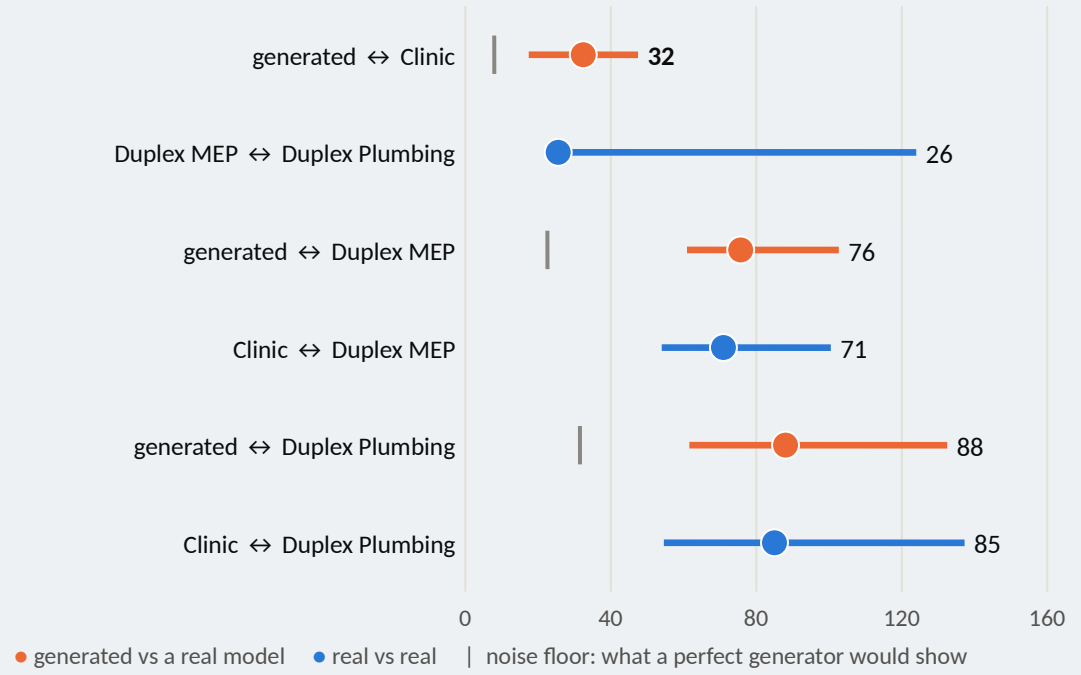
Tighter with the tier: median closest clear gap (mm)



A check that the dataset behaves as designed, not a discovery: the tier fixes the element count; load and congestion follow from the rules. Within a tier, kinds, trades, surfaces and stacking all vary. Load rises with the tier (rank correlation +0.46 on the benchmark, 125 contexts per tier; the population medians rise at every step) and the closest gap narrows at every step (−0.38); intervals are bootstrapped over contexts.

Generated spacing matches a clinic the generator never saw to within 32 millimetres





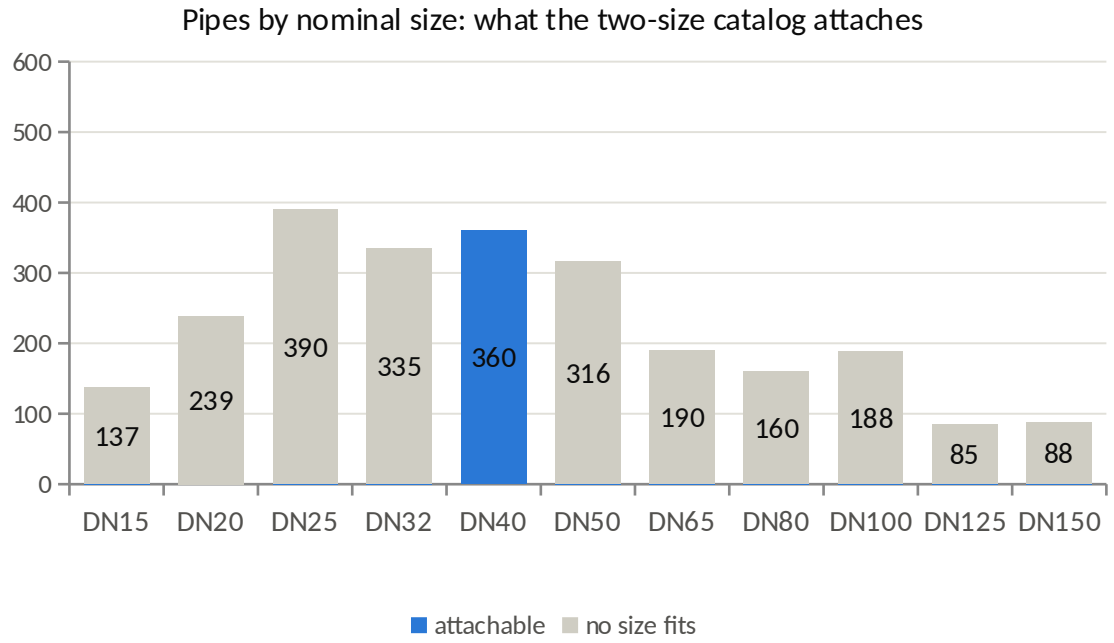
32 mm
generator to clinic (95 % CI 17–48); fitted on the duplex, clinic held out

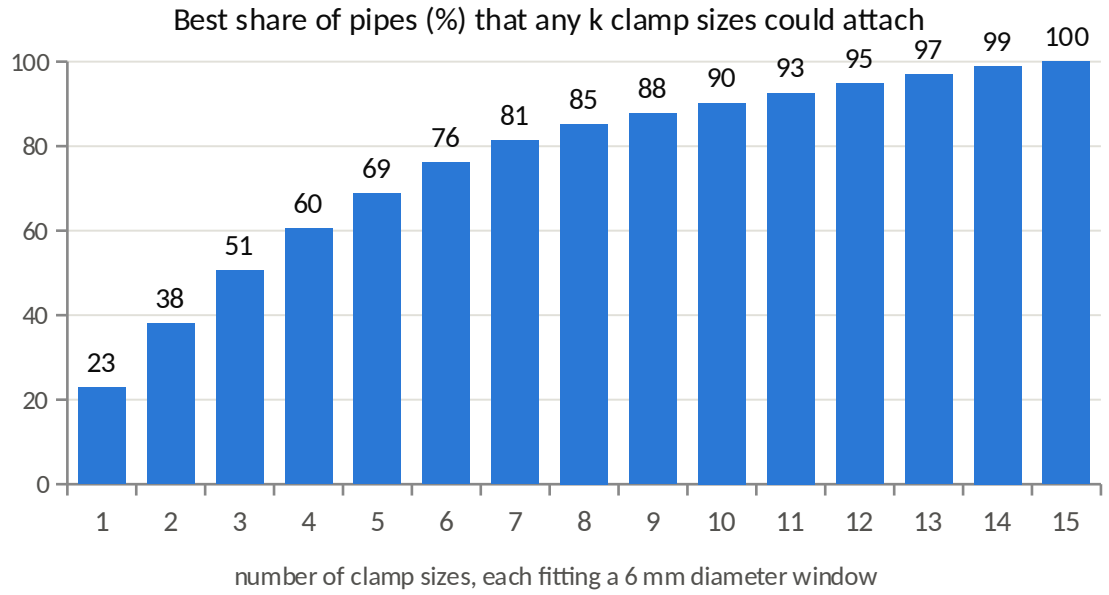
26 mm
the duplex's own two models from each other: about as close as real is to real

190 mm
what a fixed 25 mm gap would be from the clinic

Sections every 250 mm, clinic pipe pairs weighted by shared length; models: buildingSMART Duplex Apartment and Medical-Dental Clinic [12]. [12] buildingSMART International (2020), Duplex Apartment and Medical-Dental Clinic sample IFC files, CC BY 4.0

The paper’s two clamp sizes reach one pipe in seven, and the dataset says which sizes to add





14.5 %

of benchmark pipes attachable with the paper’s two sizes [2] (95 % CI 12.7–16.2); every one is DN40

38.1 %

with two sizes placed where the pipes are; 76 % with six, and every pipe with 15

0 of 2,012

trays, ducts and conduits: a pipe-clamp catalog defines no attachment. Load never binds (1.97 vs 2.5 kN)

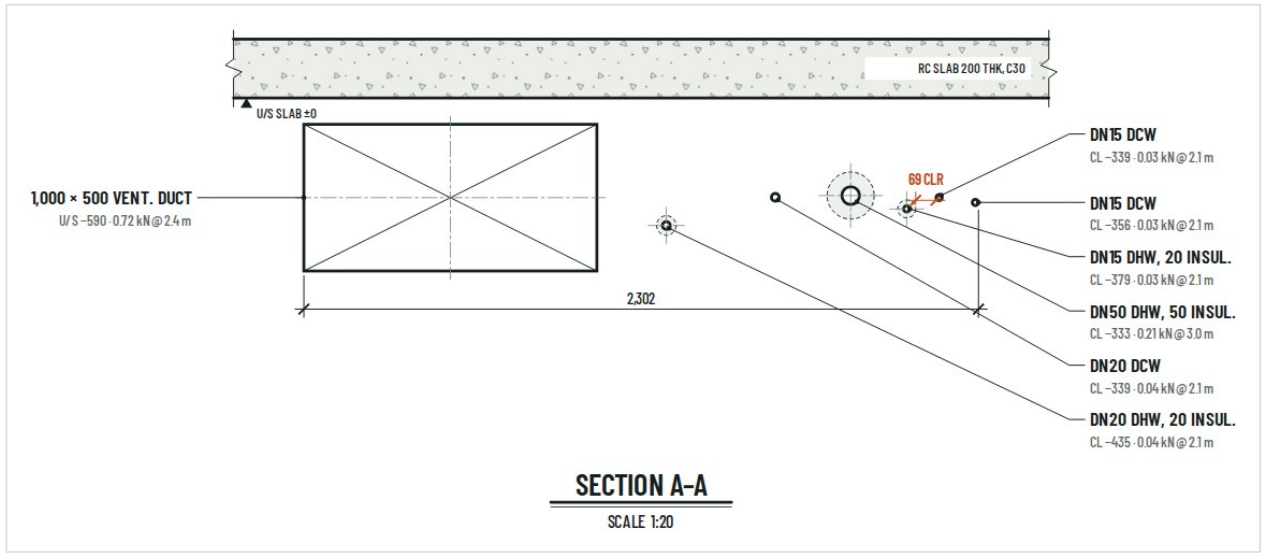
[2] Pedrollo, Gvadzabia, Graeber & Fischer (2026), CrossMEP, CIB W78 2026 (this paper)

Pick a tier or an exact mix and a seed, and the Generator Studio returns the designer’s brief

In: the parameters

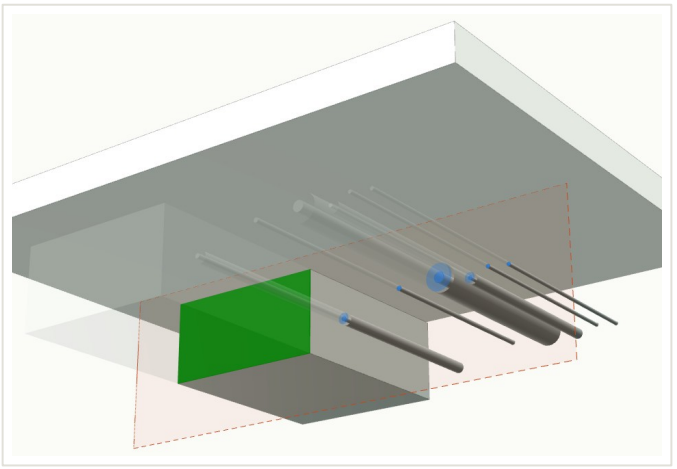


Out: section A–A as the designer receives it, and what it carries



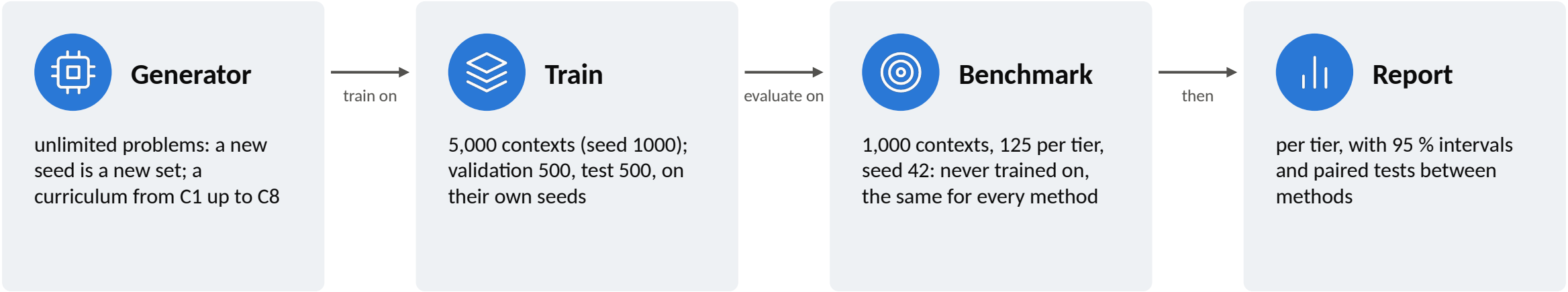
7 services in **1** row **1.10 kN** at the support closest clear gap **69 mm** hung from a **200 mm slab**

Out: the same run in 3D, cut at the section



Scan to open the studio
every section is a stored output of the released generator, with the call that reproduces it

Methods train on the generator and are compared on one fixed benchmark, tier by tier




```
import crossmep.tasks as cm
data = cm.load("benchmark")  # 1,000 contexts, or generate your own mix
from crossmep.evaluate import score, compare
score(my_results, data)  # per tier, 95 % CI    compare(mine, baseline, data)  # paired test
```

Four lines to load, score and compare; loading, metrics and scoring need only the Python standard library, the generator needs NumPy. The same files serve reinforcement learning, constraint programming and benchmarking: there are no labels to fit a method to.

Today CrossMEP is one section at one support, and next come a checker and real projects

Scope

- One cross-section at one support: routing, branches and support spacing are outside it; the span is recorded, so a method can vary it
- Realism checked on two open buildings, a residential duplex and a medical clinic; congested racks rest on practice and standards
- Trade mix and tier composition are design choices, not survey results
- Synthetic: not for the structural design of real installations

Next

1. A public checker for support designs, with best-known costs from an exact solver on the small tiers
2. The catalog as an input: generated catalogs, and catalogs a method has never seen
3. A real test set: sections from commercial projects with supports designed by engineers

If you coordinate services or design supports, open the studio, generate the sections you know from your projects, and tell us what looks wrong. Every rule is one documented constant away from being changed.

Support design finally has open problems to learn from

CrossMEP is the brief, not the answer: seven thousand support-design problems, open to anyone who wants to teach a machine to design supports.

Data CC BY 4.0, code MIT. Not for the structural design of real installations.

7,000
support-design problems in eight tiers

Sourced
every constant sourced or declared

Checked
pipe gaps tested on two open buildings

Open
data, code and generator

github.com/lavinia-ped/CrossMEP_Dataset

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